DSA REFERENCE

Depth-First Search

Recognition cues, invariants, code templates, and 5-8 referenced problems.

Explore one branch to completion while carrying the state needed to return and try alternatives.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

DSA / DFS

DSA

Depth-First Search

CORE IDEA

Explore one branch to completion while carrying the state needed to return and try alternatives.

Recognition cues

+ connected components
+ tree path properties
+ reachability
+ recursive structure

Implementation sequence

+ define visited timing
+ identify base cases
+ separate global from path-local state
+ consider recursion depth

Invariant

Every active call represents one node plus the exact path or partial state leading to it.

Worked reasoning

To count islands, start DFS at each unvisited land cell and mark the whole component before continuing the scan.

Complexity

O(V + E) for graph traversal and O(height) to O(V) stack space.

Failure modes

- [ ] marking visited too late
- [ ] reusing path state across branches
- [ ] stack overflow on deep graphs
- [ ] forgetting disconnected components

Practice ladder

Depth-First Search: referenced problems

Open the official problem page, classify the pattern, write the invariant, and predict the complexity before coding.

#	Problem	Deliberate-practice prompt
1	**Maximum Depth of Binary Tree**	State the cue, invariant, edge case, and proof before implementation.
2	**Number of Islands**	State the cue, invariant, edge case, and proof before implementation.
3	**Clone Graph**	State the cue, invariant, edge case, and proof before implementation.
4	**Path Sum**	State the cue, invariant, edge case, and proof before implementation.
5	**Surrounded Regions**	State the cue, invariant, edge case, and proof before implementation.

Before coding

+ Pattern cue
+ State
+ Invariant
+ Boundary convention

Before finishing

+ Dry run
+ Edge case
+ Time proof
+ Space proof

PUBLISHING STATUS
REVIEW GATE: STABLE TECHNICAL. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.